GERMÁN KÜBER

Senior Rust Engineer — Distributed Systems, Async Runtimes & Low-Latency Infrastructure

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PROFESSIONAL SUMMARY

Senior Rust engineer with 4+ years shipping production tokio-based async Rust in low-latency, high-throughput distributed systems, on top of 15+ years building systems across multiple languages. Designed, shipped, and operated tokio async pipelines processing $50M+ in daily transactional flow with execution under 10ms at Borderless Capital, delivering memory safety and zero-cost abstractions without garbage collection overhead. Strong systems fundamentals across async runtime patterns (tokio runtime tuning, Pin, structured concurrency), low-latency observability, Python / Cython FFI bindings, custom protocol implementations, and reverse-engineering of Go (Geth) internals to inform Rust-side designs. Distributed systems experience spans multi-chain blockchain ecosystems (EVM, Solana, Polkadot / Substrate), consensus protocols (hands-on with GRANDPA / BABE, familiar with Raft), and Zero-Knowledge proof toolchains. Comfortable in adjacent ecosystems (**TypeScript / Node.js, Python, .NET**) and across cloud-native infrastructure on AWS and Azure. Active contributor to the broader systems and Web3 communities through 80+ international talks.

CORE EXPERTISE

Rust Core & Ecosystem: Rust (stable, async / await, Pin, Generic Associated Types / GATs, async traits, trait objects, lifetimes, no_std familiar), Cargo workspace management, rustfmt, clippy, criterion (benchmarking), cargo-flamegraph (profiling), cargo-fuzz / libfuzzer (fuzzing), miri (UB detection), serde, thiserror, anyhow, tracing, tracing-subscriber

Async Runtimes & Concurrency: tokio (async runtime, runtime tuning, multi-threaded scheduler, channels, tasks, select!, spawn_blocking), futures, async-trait, pin-project, lock-free data structures (crossbeam, atomics, memory ordering), data parallelism (rayon), structured concurrency patterns, custom executors

Systems Programming & Performance: Memory safety without garbage collection, zero-cost abstractions, ownership and lifetimes, low-latency design (sub-10ms execution), FFI with C / C++ / Python, Cython bindings, custom protocol implementations, performance profiling (flamegraphs, perf), memory profiling, lock-free patterns, cache-friendly data layout

Distributed Systems & Consensus: Async-driven high-throughput pipelines, event-driven architectures, fault tolerance, distributed coordination across multi-region systems, replication patterns, eventual consistency, transaction delivery infrastructure, consensus protocols (hands-on with GRANDPA / BABE on Substrate, familiar with Raft and Paxos)

Networking & Protocols: tokio networking (TCP / UDP), HTTP servers (hyper), gRPC (tonic), custom protocol design, mempool analysis, RPC clients, TLS, parallel broadcasting across endpoints, validator / TPU routing, Linux internals (networking stack, sockets, namespaces, epoll)

Observability & Tracing: Prometheus + custom Rust telemetry, tracing crate with structured logging, OpenTelemetry, distributed tracing across services, real-time metrics design at scale (50+ production metrics)

Multi-Language Interop: Rust ↔ Python via Cython / PyO3 bindings, Rust ↔ C / C++ FFI, reverse-engineering Go (Geth / Go-Ethereum) internals, TypeScript / Node.js and .NET / C# integration, polyglot service architectures

Blockchain & Web3: Polkadot SDK, Substrate, FRAME, XCM, parachains, EVM (Ethereum, Arbitrum, Optimism), Solana, MEV infrastructure, Zero-Knowledge proofs (Noir, Barretenberg, Garaga on StarkNet, Cairo)

Cloud & Infrastructure: AWS, Azure, Docker, Kubernetes, FastAPI, CI / CD pipelines, Linux internals (networking, sockets, namespaces), Infrastructure as Code (Terraform, ARM templates)

AI / LLM Systems: LLM orchestration (multi-provider), real-time pipelines, MCP (Model Context Protocol) server and client implementation, LangChain, LangGraph

Engineering Leadership: Technical leadership for systems and Rust teams, code review and mentoring, technical interviewing, ADRs, on-call and incident response

PROFESSIONAL EXPERIENCE

Lead Architect & Co-Founder

KliverAI (Own Venture, closed Apr 2026) — Apr 2025 – Apr 2026 (full-time) — Remote, Spain

- Architected polyglot platform end-to-end (Rust, Node.js / TypeScript, Python, .NET, Kubernetes, AWS) with multi-provider service orchestration; owned ADRs, service boundaries, and technology selection.
- Designed event-driven real-time pipelines integrating multiple streaming providers with sub-200ms latency across concurrent user sessions; implemented performance-critical components in Rust (tokio, async) for the audio and streaming layer.
- Implemented MCP (Model Context Protocol) servers and clients to standardize service-to-service tool integration patterns.
- Established observability stack with OpenTelemetry, distributed tracing, and structured logging across services.
- Hired and led a distributed team of up to 3 engineers; owned technical interviews and architectural reviews.

Fractional CTO (Part-time, hands-on)

Rocking Product — Apr 2025 – Nov 2025 (part-time, parallel to KliverAI) — Remote / Buenos Aires

- Joined as Fractional CTO; led technical roadmap and architectural direction for an early-stage product, partnering directly with founders to translate business priorities into system design.
- Owned architectural design end-to-end — service boundaries, data model design, technology selection, and integration patterns — formalized through ADRs (Architecture Decision Records).
- Designed new system components and led refactoring strategy for legacy modules, balancing technical debt reduction with feature delivery velocity.
- Inherited an 8-person engineering team; established code review standards, design review process, CI / CD pipelines, and engineering rituals while remaining hands-on in the codebase.
- Optimized cloud infrastructure and runtime systems through caching, compression, and dynamic resource selection, cutting combined spend by 45%.

Lead Rust Engineer (Distributed Systems)
Borderless Capital / CTF Capital — Aug 2021 – Apr 2025 — Remote

- Shipped and operated production Rust services (tokio, async / await, Pin) processing $50M+ in daily transactional flow with execution under 10ms p99 latency; tuned tokio runtime parameters, async boundaries, and lock-free coordination patterns for sub-10ms targets.
- Built custom transaction delivery infrastructure in Rust with parallel broadcasting across multiple RPC endpoints and direct validator / TPU routes, achieving sub-slot inclusion latencies (~400ms on Solana).
- Designed async coordination across multi-chain ecosystems (EVM — Ethereum, Arbitrum, Optimism — and Solana) with custom protocol implementations, mempool analysis, and gas estimation logic.
- Shipped Python / Cython FFI bindings exposing Rust performance-critical code to higher-level orchestration layers, eliminating GC pauses across language boundaries via zero-cost abstractions.
- Engineered low-latency observability with Prometheus and custom Rust telemetry tracking 50+ real-time metrics, with structured tracing via the tracing crate across slot timing, validator rotation, and execution outcomes.
- Reverse-engineered Geth (Go-Ethereum) internals to inform Rust-side designs for transaction simulation, gas estimation, and internal trading tools.
- Researched and prototyped Zero-Knowledge proof technologies (Noir, Barretenberg, StarkNet, Cairo, account abstraction) for verifiable computation systems.
- Owned 24/7 production reliability and incident response, including on-call rotation, post-mortems, and observability ownership across trading-critical pipelines. Acted as technical lead: code review, mentoring, technical interviewing, and architectural decisions for Rust and distributed systems hires.

Senior Software Architect
Paramo Technology — Aug 2020 – Aug 2021 — Remote / Argentina

- Owned end-to-end architecture for a large-scale online platform with event-driven microservices, real-time state management, and strict consistency requirements.
- Led technical interviewing and hiring across the engineering organization; delivered internal training programs on architecture patterns and operational best practices.

Software Architect (Full-Stack)
CloudX (Software Consultancy) — Aug 2017 – Aug 2020 — Buenos Aires

- Led architecture decisions across multiple enterprise product lines, designing microservices backends using Domain-Driven Design.
- Delivered technical training and conference talks on software architecture patterns and software design.

Software Architect (Angular / .NET)
Majestic Hotel Group — Jan 2014 – Aug 2017 — Argentina

- Designed and shipped full-stack features for internal hotel management platforms; led migration from AngularJS to Angular 4 and refactored backend toward an event-oriented architecture.

.NET Web Developer
Plus54 — Aug 2009 – Mar 2014 — Buenos Aires

- Early career: built .NET web applications, internal tooling, and mobile components end-to-end.

KEY ACHIEVEMENTS

Production Distributed Systems at Borderless Capital — Shipped production tokio-based async Rust pipelines processing $50M+ in daily transactional flow with execution under 10ms across multi-chain ecosystems.
Polkadot Builder Program Graduate — Completed Polkadot Blockchain Academy in Lisbon, Portugal (2026), covering Substrate, FRAME, XCM, and runtime internals.
80+ International Technical Talks — Conferences and meetups across LATAM and Europe on Rust, distributed systems, blockchain, and software architecture.
Net-Baires Founder — Founded and grew Argentina's largest .NET developer community to 100K+ members across 10+ years.
Microsoft MVP, Developer Technologies — 8 consecutive years (2017–present), recognized for technical content and community contributions.
Microsoft Student Partner — 5 years (2016–2021).

EDUCATION & CERTIFICATIONS

Polkadot Builder Program — Polkadot Blockchain Academy, Lisbon, Portugal (2026)
Systems Engineer — Universidad Abierta Interamericana (UAI), Buenos Aires, Argentina (2017–2021)
Microsoft MVP, Developer Technologies — Feb 2017 – Present (8 consecutive years)

OPEN SOURCE & COMMUNITY LEADERSHIP

Net-Baires (.NET Community) — Founder (2015–present). Argentina's largest .NET developer community, 100K+ members, 10+ years of monthly meetups and conferences.
Crypto Para Mortales — Founder (2023–present). Spanish-language blockchain and crypto education conference.
Patrones Para Mortales — Founder (2020–present). Educational content brand on software design patterns and architecture.
YouTube channel (youtube.com/@germankuber) — technical deep-dives on Rust, blockchain, AI, and software architecture.

SPOKEN LANGUAGES

Spanish: Native — English: Full Professional Proficiency — daily use in international remote teams